Functional Annotation of *sad-II* (PP_3151, UniProt Q88I50)

NAD^+-dependent succinic semialdehyde dehydrogenase of *Pseudomonas putida* KT2440

1. Summary (Answer to the Research Question)

sad-II (locus **PP_3151**; UniProt **Q88I50**) of *Pseudomonas putida* KT2440 encodes a **NAD^+-dependent succinic semialdehyde dehydrogenase (SSADH; EC 1.2.1.24)**. Its primary, experimentally-supported (by homology) function is to catalyze the essentially irreversible, NAD^+-coupled oxidation of **succinic semialdehyde (SSA) → succinate**. This is the **terminal step of the γ-aminobutyrate (GABA) shunt / polyamine-catabolic route**, the reaction that links the degradation of GABA and biogenic amines (e.g., putrescine, spermidine) to the **tricarboxylic acid (TCA) cycle**. The enzyme is a **soluble cytoplasmic** aldehyde dehydrogenase that uses a **catalytic cysteine nucleophile** and NAD^+ hydride acceptor. It belongs to the **Sad/YneI (NAD^+-preferring) class** of bacterial SSADHs, distinct from the $NADP^+$-preferring GabD class.

2. Gene / Protein Identity Verification

Attribute	Value	Source
UniProt accession	Q88I50	UniProtKB
Gene name	*sad-II*	EMBL AAN68759.1
Locus tag	PP_3151	KT2440 genome
Organism	*Pseudomonas putida* KT2440 (ATCC 47054 / DSM 6125)	UniProtKB
Length	461 aa	UniProtKB
EC number	1.2.1.24 (SSADH, NAD^+)	UniProtKB
Family	Aldehyde dehydrogenase superfamily	UniProtKB / InterPro
Domains	Ald_DH_N (IPR016162), Ald_DH_C (IPR016163), Ald_DH_CS_CYS (IPR016160), Aldehyde_DH_dom / PF00171	InterPro / Pfam
Functional subfamily	**GABD/Sad-like (IPR047110)**, **ALDH_GabD1-like (IPR044148)**	InterPro
GO molecular function	GO:0004777 succinate-semialdehyde dehydrogenase (NAD^+) activity; GO:0004030 aldehyde dehydrogenase [$NAD(P)^+$] activity	UniProtKB/ InterPro

Verification outcome: The gene symbol *sad-II*, EC number, organism, protein family, and all listed InterPro domains are internally consistent and match the SSADH function. This report describes the correct protein. The name "*sad*" (**s**uccinic semialdehyde **d**ehydrogenase) corresponds to the *E. coli sad/yneI* gene; the "**II**" denotes one of multiple SSADH paralogs in the KT2440 genome. *No ambiguity or misidentification was encountered.*

3. Primary Function: Reaction Catalyzed and Substrate Specificity

3.1 Reaction

> Succinic semialdehyde + NAD^+ + H_2O → succinate + NADH + 2 H^+

Succinic semialdehyde dehydrogenases "catalyze the $NAD(P)^+$-coupled oxidation of succinic semialdehyde (SSA) to succinate, the last step of the γ-aminobutyrate shunt" (de Carvalho *et al.*, 2011,

P 21303655

).

The reaction is physiologically irreversible and both **detoxifies** the reactive aldehyde SSA and **generates** the TCA-cycle intermediate succinate plus reducing equivalents (NADH).

3.2 Substrate specificity

SSADHs of this family are **highly specific for succinic semialdehyde** rather than general aldehydes, and act with micromolar affinity: - Characterized NAD^+-type SSADH (Arabidopsis, a strict structural/functional analog): "specific for succinic semialdehyde ($K_{0.5}$ = 15 µM), and exclusively used NAD^+ as a cofactor (Km = 130 ± 77 µM)" with NADH a competitive inhibitor (Busch & Fromm, 1999,

P 10517851

).

- Bacterial Sad/YneI-type homolog: "shows high activity and affinity toward succinate semialdehyde and exhibits **substrate inhibition** at concentrations of SSA higher than 0.1 mM" (Zheng *et al.*, 2013,

P 23229889

).

By inference, Q88I50 acts specifically on the 4-carbon SSA with micromolar affinity, using NAD^+, with likely substrate/product inhibition that limits flux and prevents accumulation of toxic SSA.

3.3 Cofactor specificity — why "NAD$^+$-dependent"

Bacteria typically encode **two SSADH paralogs of distinct cofactor preference**: "Genomes of many organisms including *Escherichia coli* and *Salmonella typhimurium* encode two succinate semialdehyde dehydrogenases with low sequence similarity and different cofactor preference (**YneI and GabD**)" (Zheng *et al.*, 2013, P 23229889).

The **Sad/YneI** class prefers NAD$^+$ (a conserved Lys — Lys160 in YneI — "contributes to the enzyme preference to NAD($^+$)"), whereas the **GabD** class prefers NADP$^+$; structurally, "a deletion of three amino acids in *E. coli* SSADH permits this enzyme to use NADP$^+$, whereas … the human enzyme utilises NAD$^+$" (Langendorf *et al.*, 2010, P 20174634).

Q88I50 is classified **GABD/Sad-like (IPR047110)** and annotated **NAD$^+$-dependent (EC 1.2.1.24)**, placing it in the NAD$^+$-preferring Sad branch — consistent with its "*sad*" name and distinct from the genome's NADP$^+$-preferring GabD paralogs (see §3.4).

3.4 Genomic context — two SSADH systems in KT2440 (KEGG analysis)

Genome analysis of *P. putida* KT2440 shows a clear division of labor, which explains the "**-II**" in the gene name:

Locus	Symbol	KO	Cofactor / EC	Genomic context
PP_3151	**sad-II**	**K08324**	**NAD$^+$ (EC 1.2.1.24)**	**standalone**, adjacent to **PP_3152 = yneJ** (LysR regulator)
PP_0213	gabD-I	K00135	NADP$^+$ (EC 1.2.1.79/1.2.1.16)	in a **gabD-I–gabT operon** (PP_0214 = 4-aminobutyrate aminotransferase, K14268)
PP_2488	(gabD-like)	K00135	NADP$^+$	ALDH paralog
PP_4422	(gabD-like)	K00135	NADP$^+$	ALDH paralog

- **sad-II (PP_3151)** is the *sole* NAD$^+$-preferring (Sad/YneI-type, K08324) SSADH in the genome; it is a **standalone** gene at complement(3,569,479–3,570,864), **not** co-transcribed with a *gabT* gene.

- Its immediate neighbor **PP_3152 (yneJ)** is a **LysR-family transcriptional regulator**, exactly reproducing the *E. coli* **sad(yneI)–yneJ** chromosomal arrangement. This conserved synteny independently confirms PP_3151 as the genuine **Sad/YneI ortholog** and nominates YneJ as a candidate regulator of sad-II. Coordinates confirm *sad-II* is **monocistronic** and **divergently transcribed** from *yneJ*: PP_3151 is on the minus strand (complement 3,569,479–3,570,864) while PP_3152/*yneJ* is on the plus strand (3,570,972–3,571,838), the two genes reading head-to-head across a short (~108 bp) shared intergenic region — the **canonical LysR regulatory topology** (LysR regulators are typically encoded divergently from the target they control), strengthening the case that YneJ directly regulates *sad-II* independently of the *gab* operon.
- In contrast, the classical GABA-shunt pair is **gabD-I (PP_0213, $NADP^+$-SSADH) + gabT (PP_0214, GABA aminotransferase)**, encoded in a single operon — the route that transaminates GABA to SSA and oxidises it. The existence of this separate, operon-linked $NADP^+$ system underscores that **sad-II is a distinct, independently regulated NAD^+-dependent isozyme**.

KEGG pathway membership of sad-II (PP_3151): Butanoate metabolism (ppu00650; the GABA-shunt module), Alanine/aspartate/glutamate metabolism (ppu00250; GABA metabolism), and Nicotinate/nicotinamide metabolism (NAD^+/NADH cofactor turnover).

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4. Catalytic Mechanism and Active-Site Residues (structural / bioinformatic evidence)

Sequence analysis of Q88I50 identifies the hallmark aldehyde-dehydrogenase catalytic machinery:

- **Catalytic cysteine — Cys267** in the motif `QNTGQV` `C` `AAAKR`. This aligns with the experimentally validated catalytic **Cys268 of *Salmonella* YneI**, where "the NAD^+ molecule is bound in the long channel with its nicotinamide ring positioned close to the side chain of the catalytic Cys268" (Zheng *et al.*, 2013, P 23229889).
- **Coenzyme-binding / Rossmann glutamate motif** `ELGGSDPF` (residues 233–240).
- **Aldehyde-binding motif** `MPWNFP` (residues 133–138).

- Single **Aldehyde dehydrogenase domain** spanning residues 12–456 (UniProt), i.e., an N-terminal NAD-binding (Rossmann) subdomain plus a C-terminal catalytic subdomain — the classic two-domain ALDH architecture with the active site at the interdomain interface.

Quantitative alignment evidence (this work). Global Needleman–Wunsch (BLOSUM62) alignment of Q88I50 against experimentally characterized SSADHs shows it is **~2× more similar to the Sad/YneI class than to GabD**: 64.3% identity to *Salmonella* YneI (Q8ZPI3, the enzyme crystallized by Zheng *et al.* 2013), 66.3% to *E. coli* Sad/YneI (P76149), but only 35.0% to *E. coli* GabD (P25526). Critically, **every functionally validated YneI residue is conserved at the aligned position in Q88I50**:

YneI (Zheng 2013)	Role	Aligned residue in Q88I50
Cys268	Catalytic nucleophile	**Cys267** (conserved)
Trp136	Active-site / substrate	**Trp135** (conserved)
Glu365	General base (deacylation)	**Glu364** (conserved)
Asp426	Active site	**Asp425** (conserved)
Lys160	**NAD^+ vs $NADP^+$ preference determinant**	**Lys159** (conserved)

The perfect conservation of the catalytic tetrad **and** of the Lys that "contributes to the enzyme preference to NAD(+)" (Zheng *et al.*, 2013, P 23229889) provides residue-level structural evidence that Q88I50 is a genuine **NAD^+-preferring Sad/YneI-type SSADH**, rationalizing its EC 1.2.1.24 annotation.

Chemical mechanism (established for the close homolog GabD1 and shared by the family): the Cys thiol performs nucleophilic addition to the SSA carbonyl to form a **thiohemiacetal**; a **hydride is transferred to NAD^+** to give a covalent **thioacyl-enzyme** intermediate; **thioester hydrolysis** then releases succinate. "Nucleophilic addition to SSA is very fast, followed by a modestly rate-limiting hydride transfer and fast thioester hydrolysis," with product dissociation/conformational change being rate-limiting (de Carvalho *et al.*, 2011, P 21303655). Family SSADHs are homotetramers.

5. Biological Pathway Context

5.1 The GABA shunt / biogenic-amine catabolic funnel

SSADH provides the **catabolic exit** of the two-step GABA shunt:

1. GABA + 2-oxoglutarate → **succinic semialdehyde** + L-glutamate (GABA aminotransferase, GabT)
2. **Succinic semialdehyde** + NAD^+ → **succinate** + NADH **(SSADH = Q88I50)**
3. Succinate enters the **TCA cycle**.

5.2 Relevance in *Pseudomonas putida*

P. putida is a metabolically versatile soil bacterium that degrades **biogenic amines** (polyamines such as **putrescine and spermidine**, plus other amines) as growth substrates: "*Pseudomonas* species can grow in media containing different BAs as carbon and energy sources, a reason why these bacteria are excellent models for studying such catabolic pathways" (Luengo & Olivera, 2020, P 31912965). In these routes putrescine is converted (via GABA) to succinic semialdehyde, which SSADH oxidizes to succinate — allowing the bacterium to use these nitrogenous compounds as **carbon, nitrogen and energy sources** while detoxifying the reactive aldehyde intermediate. Thus sad-II funnels the degradation of GABA and polyamines into central carbon metabolism.

5.3 Physiological significance: detoxification + redox + anaplerosis

The single reaction catalyzed by sad-II serves three linked physiological purposes: 1. **Aldehyde detoxification** — SSADHs are "essential for preventing accumulation of toxic levels of succinic semialdehyde (SSA) in cells" (Langendorf *et al.*, 2010, P 20174634); clearing this reactive aldehyde protects the cell. 2. **Energy conservation (redox)** — the oxidation reduces NAD^+ to **NADH**, feeding reducing equivalents into the respiratory chain. 3. **Carbon assimilation / anaplerosis** — the product **succinate** enters the TCA cycle, letting GABA/polyamine carbon be used for growth. Because NAD^+ reduction and thioester hydrolysis make the reaction thermodynamically favorable and effectively irreversible, sad-II acts as a committed, one-way catabolic exit.

5.4 Regulatory/signaling note

In some bacteria the intracellular SSA level set by SSADH activity acts as a signal (e.g., succinic semialdehyde couples stress response to quorum-sensing signal decay in *Agrobacterium tumefaciens*, where SSADH activity "might control the intracellular SSA level"; Wang *et al.*, 2006,

P 16942602

).

While such a signaling role has not been demonstrated for PP_3151 specifically, it illustrates that SSADH activity governs a metabolically important pool of SSA. In KT2440, *sad-II* is likely under control of the divergently encoded LysR regulator **YneJ (PP_3152)** (see §3.4).

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6. Subcellular Localization

Q88I50 comprises a single soluble aldehyde-dehydrogenase domain (residues 12–456) with **no signal peptide and no transmembrane segment**. Bacterial SSADHs are **cytoplasmic**; the eukaryotic orthologs are soluble matrix enzymes (e.g., plant SSADH is localized to the mitochondrial matrix; Busch & Fromm, 1999,

P 10517851

).

The enzyme therefore carries out its reaction in the **bacterial cytoplasm**, where GABA-shunt and polyamine-catabolism intermediates are generated.

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7. Supported and Refuted Hypotheses

Hypothesis	Verdict	Basis
Q88I50 catalyzes SSA → succinate (EC 1.2.1.24)	**Supported**	UniProt/GO annotation; family definition (P 21303655)
Uses a catalytic Cys nucleophile + NAD^+ hydride acceptor	**Supported**	Cys267 alignment to YneI Cys268 (P 23229889); mechanism (P 21303655)
Belongs to the NAD^+-preferring Sad/YneI class (vs. $NADP^+$ GabD)	**Supported**	InterPro GABD/Sad-like; EC 1.2.1.24; two-paralog system (PMIDs 23229889, 20174634)
Specific for succinic semialdehyde with µM affinity	**Supported (by homology)**	PMIDs 10517851, 23229889
Functions in cytoplasm as terminal step of GABA/polyamine catabolism feeding TCA	**Supported**	Pathway definition (P 21303655); Pseudomonas BA catabolism (P 31912965); no signal peptide
Q88I50 is closer to Sad/YneI than GabD at sequence level	**Supported**	Global alignment: 64–66% id to Sad/YneI vs 35% to GabD (this work)
NAD^+-specificity determinant (YneI Lys160) is conserved	**Supported**	Lys159 conserved at aligned position (this work; P 23229889)
sad-II is regulated separately from the *gab* operon (by YneJ)	**Supported (inference)**	Monocistronic gene divergent from LysR *yneJ*; distinct from gabD-I–gabT operon (KEGG genome context, this work)
Direct biochemical characterization of PP_3151 itself	**Not available**	No primary study of this exact locus found; annotation rests on strong homology

8. Limitations and Future Directions

- **No primary biochemical or structural study of PP_3151 itself** was located in the available literature; the functional assignment is a robust **inference from family membership, conserved active-site residues, and characterized orthologs** (E. coli/ Salmonella YneI & GabD, human, plant, *M. tuberculosis* GabD1).
- The **exact cofactor kinetics (NAD^+ vs $NADP^+$ Km), substrate-inhibition constant, and oligomeric state** of Q88I50 should be measured directly (recombinant expression + steady-state kinetics).
- The **paralog landscape** in KT2440 is now mapped (this work): the $NADP^+$ GabD class = PP_0213 (**gabD-I**, in a *gabD-I–gabT* operon), PP_2488, PP_4422; sad-II (PP_3151) is the lone NAD^+ K08324 isozyme. Which catabolic route (GABA vs putrescine vs 4-hydroxybutyrate/ other SSA sources) preferentially recruits sad-II, and whether YneJ (PP_3152) activates it, still require **transcriptomic/genetic confirmation** (e.g., growth and expression of ΔPP_3151 and Δ*yneJ* on GABA/putrescine).
- A crystal structure or AlphaFold model with docked NAD^+/SSA would confirm the roles of Cys267, the general-base glutamate, and the NAD^+-specificity residues predicted here.

Key References

- de Carvalho LPS *et al.* (2011) Chemical mechanism of SSADH (GabD1) from *M. tuberculosis*.

  P 21303655

- Zheng H *et al.* (2013) Structure and activity of $NAD(P)^+$-dependent SSADH YneI from *S. typhimurium*.

  P 23229889

- Langendorf CG *et al.* (2010) X-ray structure of *E. coli* GabD SSADH; $NADP^+$ interactions.

  P 20174634

- Busch KB & Fromm H (1999) Plant SSADH: purification, mitochondrial localization, NAD^+ specificity.

  P 10517851

- Luengo JM & Olivera ER (2020) Catabolism of biogenic amines in *Pseudomonas* species.

  P 31912965

- Wang C *et al.* (2006) Succinic semialdehyde couples stress response to QS signal decay in *A. tumefaciens*.

  P 16942602